eventPublish

Service Description

Service ID: *"eventPublish"*

Abstract
This document describes the `eventPublish` service, version 1.0.0, produced by the EventHandler support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `eventPublish` service, which provides publication of events to the subscribers of their event type. It is produced by the EventHandler system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 EventHandler publish service.

2.2 How This Service Is Meant to Be Used

A publisher calls `publish` with the event type, payload and metadata; the EventHandler posts the event to every matching subscriber's `notifyUri` with its own certificate and identity token.

2.3 Important Delimitations and Dependencies

Delivery is asynchronous and best effort; the result only counts matched subscriptions and accepted deliveries. With `service.authorization.enabled=true` publishing needs a grant.

3 Service Interface

This section describes the interfaces to the `eventPublish` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `eventPublish` service and its operations.

The following interface operations are available.

3.1 operation publish (EventPublishRequest) : EventPublishResult

Publishes an event to the matching subscriptions.

4 Information Model

Here, all data objects that can be part of the `eventPublish` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.3, which are used to represent things like hashes and identifiers.

4.1 struct EventPublishRequest

An event to publish.

Field	Type	Description
eventType	String	Event type, camelCase.
payload	String	Event payload (text).
metadata	Metadata	Event metadata, matched against subscription filters.
timestamp	String	Time of the event (RFC 3339).

4.2 struct EventPublishResult

Outcome of a publication.

Field	Type	Description
matchedSubscriptions	Number	Number of subscriptions the event matched.
acceptedDeliveries	Number	Number of notifications accepted for delivery.

4.3 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “EventHandler – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-EventHandler-java-spring/documentation/v5.3.0/EventHandler-v5.3.0-SysD.pdf.

[2] ——, “eventPublish HTTP/TLS/JSON – Interface Design Description (EventHandler),” Eclipse Arrowhead, IDD 5.3.0, 2026, eventHandler-v5.3.0-eventPublish-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	